How Many Scales Can You Build?

Dimensional Analysis, Gaussian Elimination, and the Geometry of Mass and Entropy

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Abstract

How many independent mass scales can you construct from the fundamental constants of physics? We develop this question from first principles, beginning with Bridgman's axiom and the Buckingham π theorem, through warm-up problems (the spring period, the density of the universe) to the full six-constant problem. Writing "mass equals some product of constants" is equivalent to solving a system of four linear equations in six unknowns. Gaussian elimination, presented step by step as a teaching example, reveals rank four and nullity two: a *two-parameter family* of dimensionally consistent mass scales, $m(s,t) = c^{-2-5s}\, h^{1+s-t}\, H^{1+2s-t}\, G^{s}\, (kT)^{t}$. The Planck mass, the Hubble mass, and Weinberg-type scales appear as named points in a continuous (s,t)-plane, and the geometric mean structure of the exponent space reveals that $m_P = \sqrt{m_H \cdot m_\gamma}$.

Dimensional analysis forces Boltzmann's constant and temperature into an inseparable pair (kT): a structural lock that is derived, not assumed. Following this thermal direction off the equator leads to entropy. The ratio $kT/(hH)$ that parameterises the t-axis generates a dimensionless action measure $\Sigma_{MI} = Mc^2/(hH) \sim 10^{120}$, while the area ratio R_H^2/ℓ_P^2 produces the horizon entropy $S_{dS}/k_B \sim c^5/(\hbar G H^2) \sim 10^{122}$. These two entropies scale differently with the Hubble parameter (H^{-1} versus H^{-2}), reflecting distinct geometric origins: volume–time action versus horizon area. Dimensional analysis fixes the scalings; dynamics supplies the geometric prefactors. Finally, we identify the two dials themselves: the s-axis is the cosmological hierarchy $\Pi_0 = hGH^2/c^5 \approx 10^{-122}$ and the t-axis is the thermal ratio $kT/(hH)$, so that the 121-order span of the equator is Π_0^{-1} rather than a coincidence. Admitting the elementary charge adds a third dial, the fine-structure constant, recovering as a limiting case the result that gravity enters no dimensionless combination with c, $\hbar$ and e. Because no particle mass lies in the plane, the classical large-number coincidences require a further invariant $\alpha_G = Gm_p^2/\hbar c$; in that basis Dirac's hypothesis reads $\alpha_G^2 \sim \Pi_0$ and fails by 45 orders of magnitude.

Contents

Foundations: Why Dimensional Analysis Works

Every physics student meets $m_P = \sqrt{hc/G}$ and is told "this is the Planck mass." But where does it come from? Is it the *only* mass you can build from fundamental constants? And what happens when you throw in the Hubble constant, Boltzmann's constant, and a temperature?

Suppose you are given six constants:

$$c, \quad h, \quad G, \quad H_0, \quad k, \quad T.$$

Speed of light, Planck's constant, Newton's gravitational constant, the Hubble parameter, Boltzmann's constant, and a cosmological temperature. You want to combine them into a quantity with the dimensions of **mass**. How many independent ways can this be done? This is merely a question of **linear algebra**. The constraints come from balancing dimensions, and the answer, as we shall see, is not a single mass but a **two-dimensional family**. We will build to this result in stages, starting from the formal foundations.

Before we use dimensional analysis, we should understand why it is legitimate. The answer rests on a single axiom and a single theorem.

Bridgman's axiom

Percy Bridgman articulated the principle that underlies all dimensional reasoning [1]:

The form of a physical law must be invariant under changes of scale of fundamental units.

If we measure length in metres or feet, the *form* of Newton's second law $F = ma$ does not change; only the numerical values of the constants adjust. This axiom is a statement about what we mean by a "physical law." It rules out equations like $F = 3.7\, m + a$, which would change form under a rescaling of units. Bridgman's axiom is what licenses us to extract structural information from units alone.

The Buckingham π theorem

Bridgman's axiom has a precise algebraic consequence [2]:

Theorem 1 (Buckingham π theorem)**.** *Any physically meaningful equation involving n variables expressible in terms of k independent fundamental dimensions can be rewritten as an equation involving $n - k$ dimensionless parameters (called π-terms).*

The number k is the **rank** of the dimensional matrix, that is, the matrix whose rows are the fundamental dimensions and whose columns are the variables. The quantity $n - k$ is the **nullity**: the number of free parameters, the dimension of the space of dimensionless combinations. This is rank–nullity from linear algebra, applied to physics.

Warm-Up: The Period of a Spring

We start with a problem where the answer is well known. Consider a mass m on a spring with constant κ, displaced by l. We seek an expression for the system's period $\mathcal{T}$. The variables and their dimensions:

	$\mathcal{T}$	m	κ	l
M	0	1	1	0
L	0	0	0	1
T	1	0	-2	0

We have $n = 4$ variables and rank $k = 3$ (all rows are independent). By the π theorem, there is $4 - 3 = 1$ dimensionless combination:

$$\pi_1 = \mathcal{T}^a \, m^b \, \kappa^c \, l^d.$$

Requiring $[\pi_1] = M^0 L^0 T^0$ gives $b + c = 0$, $d = 0$, $a - 2c = 0$. Choosing $c = 1$: $b = -1$, $a = 2$, $d = 0$. So

$$\pi_1 = \frac{\kappa \mathcal{T}^2}{m} \qquad \Longrightarrow \qquad \mathcal{T} \sim \sqrt{\frac{m}{\kappa}},$$

which matches the known result $\mathcal{T} = 2\pi\sqrt{m/\kappa}$. The dimensionless factor 2π is invisible to dimensional analysis; it requires dynamics to determine.

Checkpoint

Lesson 1: The displacement l drops out entirely. Dimensional analysis tells us the period *cannot* depend on amplitude, a non-trivial physical result obtained without solving any differential equation.
Lesson 2: With one π-term, the system is *fully determined* up to a dimensionless constant. There is no free parameter.

Mass Scales from Six Constants

Before tackling the full six-constant mass problem, we apply the method to a cosmological question that is still fully determined: what is the density of the universe, given G, H, and c? The density ρ has dimensions $[ML^{-3}]$. With G, H, and c we have $n = 4$ variables. The dimensional matrix:

	G	H	c	ρ
M	-1	0	0	1
L	3	0	1	-3
T	-2	-1	-1	0

Rank $k = 3$, so there is $4 - 3 = 1$ dimensionless π-term. Writing $\pi = G^\alpha H^\beta c^\gamma \rho$ and solving:

$$M: \; -\alpha + 1 = 0, \qquad L: \; 3\alpha + \gamma - 3 = 0, \qquad T: \; -2\alpha - \beta - \gamma = 0.$$

From M: $\alpha = 1$. Then $\gamma = 0$ and $\beta = -2$. So

$$\pi = \frac{G\rho}{H^2} \qquad \Longrightarrow \qquad \rho \sim \frac{H^2}{G}.$$

The Friedmann equation for a flat universe gives $\rho = 3H^2/(8\pi G)$, with the factor $3/(8\pi)$ arising from the specific structure of the Einstein tensor in the FLRW metric.

Checkpoint

Lesson 3: Again fully determined: one π-term, no free parameter. The speed of light c drops out ($\gamma = 0$), meaning the critical density depends only on G and H. This is a structural result, not an approximation.
Lesson 4: Adding more constants (like h) to this problem *increases* the nullity without changing the rank, creating free parameters, which is what happens next.

We now tackle the full problem. With six constants and four base dimensions, Buckingham predicts at least two free parameters so the question becomes: what does that freedom look like? As such we seek mass scales of the form

$$m \sim c^{\alpha}\, h^{\beta}\, H^{\gamma}\, G^{\delta}\, k^{\varepsilon}\, T^{\eta}, \tag{1}$$

where each constant has known dimensional content:

Constant	Symbol	Dimensions
Speed of light	c	LT^{-1}
Planck's constant	h	ML^2T^{-1}
Gravitational constant	G	$M^{-1}L^3T^{-2}$
Hubble parameter	H	T^{-1}
Boltzmann constant	k	$ML^2T^{-2}\Theta^{-1}$
Temperature	T_{cmb}	Θ

Requiring $[m] = M^1L^0T^0\Theta^0$ gives four equations in six unknowns:

$$\begin{aligned}
M&:\quad \beta - \delta + \varepsilon = 1,\\
L&:\quad \alpha + 2\beta + 3\delta + 2\varepsilon = 0,\\
T&:\quad -\alpha - \beta - \gamma - 2\delta - 2\varepsilon = 0,\\
\Theta&:\quad -\varepsilon + \eta = 0.
\end{aligned} \tag{2}$$

Checkpoint

Count: four equations, six unknowns. By rank–nullity, the solution space has dimension $\geq 6 - 4 = 2$. A word on how this compares with the earlier problems, since the bookkeeping differs. There, the target quantity ($\mathcal{T}$, then ρ) was included as a column and we sought π-terms: the nullspace was one-dimensional, and that single direction *was* the answer, fixed up to a dimensionless constant by setting the target's own exponent to unity. The result was fully determined, with no free parameter. Here the target sits on the right-hand side instead, and we solve an inhomogeneous system; its solutions are one particular answer. The two dimensions of that nullspace are therefore genuine freedoms, and the contrast is between **zero** free parameters before and **two** now. Adding k and T to the constant pool is what creates them.

The First Surprise: The kT Lock

Before performing Gaussian elimination, we can read one equation directly. This is a lesson in inspecting a system before solving it: sometimes one row is already reduced.

The fourth equation in (2) states:

$$-\varepsilon + \eta = 0 \qquad \Longrightarrow \qquad \eta = \varepsilon.$$

The exponent of T must equal the exponent of k. They cannot appear independently:

$$k^{\varepsilon} T^{\eta} = (kT)^{\varepsilon}.$$

Key insight

Dimensional analysis **forces** k and T into the product kT, the thermal energy scale. This is not assumed; it is a consequence of a single row of the dimensional matrix. The Boltzmann distribution, equipartition, blackbody radiation: they all use kT. Dimensional analysis doesn't know any of that physics, but arrives at the same conclusion.

Gaussian Elimination: The Main Act

This section is the pedagogical heart of the paper. We perform Gaussian elimination on a 4×6 augmented matrix, narrating every row operation with its geometric and physical meaning.

Step 0: Initial matrix

The augmented matrix with columns $(\alpha, \beta, \gamma, \delta, \varepsilon, \eta \,|\, \text{RHS})$:

$$\begin{array}{cc} \left[\begin{array}{cccccc|c} 0 & 1 & 0 & -1 & 1 & 0 & 1 \\ 1 & 2 & 0 & 3 & 2 & 0 & 0 \\ -1 & -1 & -1 & -2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 \end{array}\right] & \begin{array}{l} \leftarrow M \\ \leftarrow L \\ \leftarrow T \\ \leftarrow \Theta \end{array} \end{array} \tag{3}$$

Reading the matrix. Row 1 is mass (M), Row 2 is length (L), Row 3 is time (T), Row 4 is temperature (Θ). The columns correspond to $\alpha, \beta, \gamma, \delta, \varepsilon, \eta$. Row 1 has zero in α-column: speed of light carries no mass dimension. We cannot pivot on it, so we swap.

Step 1: Swap $R_1 \leftrightarrow R_2$

$$\left[\begin{array}{cccccc|c} 1 & 2 & 0 & 3 & 2 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 & 1 \\ -1 & -1 & -1 & -2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 \end{array}\right] \tag{4}$$

Geometric meaning. Row swaps reorder equations without changing the solution. We place the length constraint first because it involves all the "spatial" constants.

Step 2: $R_3 \to R_3 + R_1$

$$\left[\begin{array}{cccccc|c} \mathbf{1} & 2 & 0 & 3 & 2 & 0 & 0 \\ 0 & \mathbf{1} & 0 & -1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 \end{array}\right] \tag{5}$$

What happened. Adding length to time eliminates α from Row 3. Physically: we combined the L and T constraints into a statement about h, G, H, and k only, projecting out one dimension of exponent space.

Step 3: $R_3 \to R_3 - R_2$

$$\left[\begin{array}{cccccc|c} \mathbf{1} & 2 & 0 & 3 & 2 & 0 & 0 \\ 0 & \mathbf{1} & 0 & -1 & 1 & 0 & 1 \\ 0 & 0 & \mathbf{-1} & 2 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 \end{array}\right] \tag{6}$$

Step 4: $R_3 \to -R_3, \quad R_4 \to -R_4$

$$\left[\begin{array}{cccccc|c} \mathbf{1} & 2 & 0 & 3 & 2 & 0 & 0 \\ 0 & \mathbf{1} & 0 & -1 & 1 & 0 & 1 \\ 0 & 0 & \mathbf{1} & -2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & \mathbf{1} & -1 & 0 \end{array}\right] \tag{7}$$

Checkpoint

Row echelon form achieved. Pivots in columns 1, 2, 3, 5 (corresponding to α, β, γ, ε). Free variables in columns 4 and 6 (δ and η). Rank = 4, nullity = 2.

Step 5: Back-substitution

We name the free parameters:

$$s := \delta \quad \text{(gravity dial)}, \qquad t := \varepsilon = \eta \quad \text{(thermal dial)},$$

$$\begin{array}{c} \begin{array}{cccccc} \alpha & \beta & \gamma & \delta & \varepsilon & \eta \end{array} \\ \left[\begin{array}{cccccc|c} \mathbf{1} & 2 & 0 & 3 & 2 & 0 & 0 \\ 0 & \mathbf{1} & 0 & -1 & 1 & 0 & 1 \\ 0 & 0 & \mathbf{1} & -2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & \mathbf{1} & -1 & 0 \end{array}\right] \end{array} \tag{8}$$

so that the monomial (1) becomes $m \sim c^\alpha\, h^\beta\, H^\gamma\, G^s\, (kT)^t$, with the pivot variables α, β, γ to be determined.

Row 4: $\varepsilon - \eta = 0$ confirms $\eta = t$ (the kT lock from Section 4).

Row 3: $\gamma - 2\delta + \varepsilon = 1$ gives $\gamma = 1 + 2s - t$.
Reading this: the Hubble exponent increases with gravity (s) and decreases with temperature (t). At $(s, t) = (-1, 0)$: $\gamma = -1$, recovering the Hubble mass $c^3/(GH)$.

Row 2: $\beta - \delta + \varepsilon = 1$ gives $\beta = 1 + s - t$.
Reading this: at $(s,t) = (-\frac{1}{2}, 0)$: $\beta = \frac{1}{2}$, recovering the Planck mass $\sqrt{hc/G}$.

Row 1: $\alpha + 2(1 + s - t) + 3s + 2t = 0$ gives $\alpha = -2 - 5s$.

Reading this: the speed-of-light exponent is entirely controlled by gravity. Every unit increase in s pushes α down by 5.

The general solution

$$\boxed{\begin{array}{lll} \alpha = -2 - 5s, & \beta = 1 + s - t, & \gamma = 1 + 2s - t, \\ \delta = s, & \varepsilon = t, & \eta = t. \end{array}} \tag{9}$$

$$\boxed{m(s,t) = c^{-2-5s}\, h^{1+s-t}\, H^{1+2s-t}\, G^{s}\, (kT)^{t}.} \tag{10}$$

Key insight

The answer is not a mass. It is a **two-parameter family** of masses. Every $(s,t) \in \mathbb{R}^2$ gives a dimensionally consistent mass scale. Planck, Hubble, and Weinberg are **coordinates** in this plane.

The (s,t)-Plane: A Map of Mass Scales

We now have a coordinate system on the space of all dimensionally consistent masses. Let us draw the map and locate the familiar landmarks.

The thermal equator: $t = 0$

Setting $t = 0$ eliminates thermal contributions:

$$m(s,0) = c^{-2-5s}\, h^{1+s}\, H^{1+2s}\, G^{s}.$$

Name	(s,t)	**Expression**	**Order (kg)**
Graviton scale m_γ	$(0,\, 0)$	hH/c^2	$\sim 10^{-68}$
Weinberg scale	$(-\frac{1}{3},\, 0)$	$(h^2cH/G)^{1/3}$	$\sim 10^{-66}$
Planck mass m_P	$(-\frac{1}{2},\, 0)$	$\sqrt{hc/G}$	$\sim 10^{-8}$
Hubble mass m_H	$(-1,\, 0)$	$c^3/(GH)$	$\sim 10^{52}$

Table 1: Named points on the thermal equator. Moving s from 0 to -1 sweeps **120 orders of magnitude**.

The geometric mean structure

The linearity of the exponent space has a beautiful consequence: the geometric mean of two mass scales corresponds to the *midpoint* of their (s, t) coordinates.

On the thermal equator, a mass at parameter s has exponents that depend linearly on s. The geometric mean of masses at s_1 and s_2 therefore sits at $\bar{s} = (s_1 + s_2)/2$. Consider the graviton mass $m_\gamma = hH/c^2$ at $s = 0$ and the Hubble mass $m_H = c^3/(GH)$ at $s = -1$. Their midpoint is $s = -\frac{1}{2}$:

$$\sqrt{m_\gamma \cdot m_H} = \sqrt{\frac{hH}{c^2} \cdot \frac{c^3}{GH}} = \sqrt{\frac{hc}{G}} = m_P. \tag{11}$$

Key insight

The Planck mass is the geometric mean of the graviton and Hubble masses:

$$m_P = \sqrt{m_\gamma \cdot m_H}.$$

This is a direct consequence of the linear structure of the exponent space: $s = -\frac{1}{2}$ is the midpoint of $s = 0$ and $s = -1$. Further geometric mean relationships are explored in Appendix A.

The dimensionality map

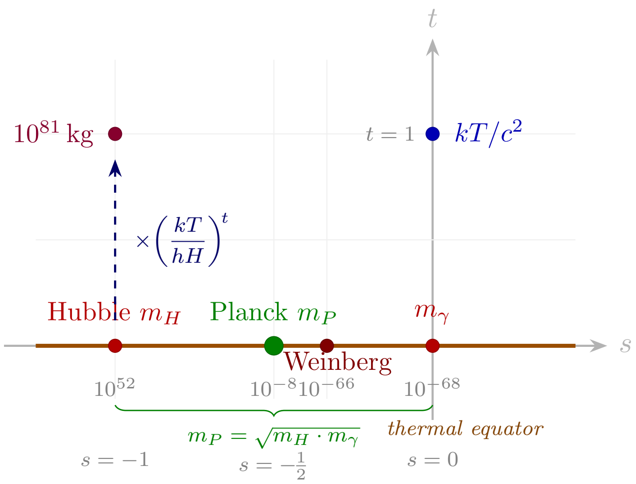

Figure 1: The (s, t)-plane of dimensionally admissible mass scales. Named masses sit on the thermal equator ($t = 0$), with orders of magnitude in grey. The Planck mass sits at the midpoint $s = -\frac{1}{2}$, the geometric mean of the Hubble and graviton scales. Moving vertically multiplies by powers of $kT/(hH) \sim 10^{29}$.

What the Dials Are: Two Invariants and a Third

We have a plane of mass scales with two dials, s and t, and we have named points on it. But we have not asked what the dials themselves *are*. The answer is that each dial is a dimensionless number of independent physical interest, and that a seventh constant would add a third dial.

The directions of the plane

The mass family (10) solves the inhomogeneous system $\mathbf{Dp} = \mathbf{t}_M$, where $\mathbf{t}_M = (1,0,0,0)^\top$ is the dimension vector of mass. The solution set of any such system is a particular solution plus the nullspace of $\mathbf{D}$, so the two free parameters are coordinates on $\ker \mathbf{D}$. It is worth asking what that nullspace contains, since by the Buckingham theorem of Section 1 its elements are precisely the π-terms of the constant set [2].

Differentiate the exponent vector (9) with respect to each free parameter. In the column order (c, h, H, G, kT), with k and T locked into the single symbol kT by Section 4,

$$\frac{\partial}{\partial s} = (-5,\ 1,\ 2,\ 1,\ 0), \qquad \frac{\partial}{\partial t} = (0,\ -1,\ -1,\ 0,\ 1).$$

Both must lie in $\ker \mathbf{D}$, since moving in either direction leaves the dimension of the result unchanged; read back as monomials they are

$$\Pi_0 = \frac{hGH^2}{c^5} \approx 8.7 \times 10^{-122}, \tag{12}$$

$$\Theta_T = \frac{kT}{hH} \approx 2.6 \times 10^{28}. \tag{13}$$

Key insight

The s-axis *is* the cosmological hierarchy $\Pi_0 = hGH^2/c^5$, and the t-axis *is* the thermal ratio $kT/(hH)$. The two dials are more than convenient parameters being the two independent dimensionless numbers this constant set admits. Sliding along s multiplies a mass by powers of Π_0; sliding along t multiplies it by powers of Θ_T. The factorisation $m(s,t) = m(s,0)\,\Theta_T^t$ of Eq. (17) was the t-half of this statement; Eq. (12) supplies the s-half.

The 121 orders, explained

This settles a number the paper has so far only recorded. The graviton mass sits at $s = 0$ and the Hubble mass at $s = -1$, so their ratio is exactly one unit of the s-dial:

$$\frac{m_H}{m_\gamma} = \frac{c^3/(GH)}{hH/c^2} = \frac{c^5}{hGH^2} = \Pi_0^{-1}, \qquad \log_{10}\Pi_0^{-1} \approx 121. \tag{14}$$

The 121-order span of Table 1 is therefore not a coincidence of the numbers but the cosmological hierarchy measured on a logarithmic ruler. The same invariant appears elsewhere in different dress: in Planck units as $H^2 \sim 10^{-122}$, in a Hubble-adapted basis $c = h = H = 1$ as $G \sim 10^{-122}$, and through the de Sitter relation $H^2 = \Lambda c^2/3$ as $\Lambda \ell_P^2 \sim 10^{-122}$, the form in which it is usually met as the cosmological constant problem [14]. Its reciprocal is the Hubble-horizon area in Planck units, $\Pi_0^{-1} = (R_H/\ell_P)^2$, which is why the horizon entropy of Section 8 carries the same magnitude.

Checkpoint

Lesson: a distance in exponent space is the logarithm of a dimensionless invariant. Every gap between named points on the equator is a power of Π_0; every vertical displacement is a power of Θ_T. The map is calibrated in units of the two invariants.

Adding charge: a third dial

What if we admit a seventh constant, the elementary charge? Taking e in Gaussian form, so that Coulomb's law reads $F = e^2/r^2$ with no separate constant and $[e] = M^{1/2}L^{3/2}T^{-1}$, the constant set becomes $\{c, h, H, G, k, T, e\}$. The dimensional matrix acquires a column but no new row, so the rank stays at four and the nullity rises from two to three. The mass family becomes a three-parameter one, and the new direction is

$$\alpha' = \frac{e^2}{hc} = \frac{\alpha}{2\pi} \approx 1.16 \times 10^{-3}, \tag{15}$$

with $\alpha = e^2/(\hbar c) \approx 1/137.036$ the fine-structure constant.

That α is the unique invariant contributed by charge is the central result of Banerjee [9], reached from the complementary direction. Working with $\{c, \hbar, G, e\}$ and no cosmological constant, he solves $[G]^{x_G}[c]^{x_c}[\hbar]^{x_h}[e]^{x_e} = M^0L^0T^0$ and finds $x_G = 0$ uniquely: the gravitational constant enters no dimensionless combination with the other three, so $\hbar c/e^2$ is the only invariant available. His conclusion, that gravity stands apart from the other domains of physics, is the $H \to 0$ corner of the present map. The same analysis leaves the exponent of h free in the definition of a natural mass, so that the Stoney and Planck unit systems are the choices $x_h = 0$ and $x_h = \frac{1}{2}$ of a one-parameter family whose members differ by powers of $\sqrt{\alpha}$; this is the $t = 0$ equator of Section 6 in a different constant basis. Whether such invariants should be counted as fundamental, and how many there are, is the subject of a long controversy [13].

Dial	Invariant	Value	Generated by
s (gravity)	$\Pi_0 = hGH^2/c^5$	8.7×10^{-122}	c, h, H, G
t (thermal)	$\Theta_T = kT/(hH)$	2.6×10^{28}	h, H, k, T
u (charge)	$\alpha' = e^2/(hc)$	1.16×10^{-3}	c, h, e

Table 2: The three invariant directions. Each is the unique dimensionless combination of its own four-constant sub-basis: dropping e and kT leaves Π_0 alone, dropping H and kT leaves α alone (Banerjee's case [9]), and the full set carries all three.

Key insight

The fine-structure constant and the cosmological hierarchy are siblings. Each is the sole invariant of a four-constant sub-basis, and each becomes an axis of the family when the sub-bases are combined. The (s, t)-plane is the α-free slice of a three-dimensional (s, t, u) space.

Why the plane contains no particle

One consequence deserves emphasis, because it constrains what the map can be asked to do. Every point of the (s, t)-plane is a mass *built from constants.* A physical particle mass is not such a point: m_e and m_p are measured parameters of the matter content, and no combination of c, h, H, G, k, T and e produces them. As Banerjee notes, the electron's charge is a quantised universal quantity whereas its mass is not, which is why particle masses are conventionally counted as parameters rather than constants [9, 13].

Locating a particle on the map therefore requires adjoining its mass as a further quantity, which raises the nullity again and introduces a fourth invariant, the gravitational coupling

$$\alpha_G = \frac{Gm_p^2}{\hbar c} = \left(\frac{m_p}{m_P}\right)^2 \approx 5.9 \times 10^{-39}. \tag{16}$$

In the language of the map, α_G is precisely the proton's displacement from the Planck point $s = -\frac{1}{2}$, expressed as a squared ratio. This is the same move made in Section 8 when the Schwarzschild and Compton lengths were introduced: a free mass parameter changes the dimensional problem. Appendix B follows the consequence for the classical large-number coincidences, which all live in the enlarged space rather than on the plane.

The Thermal Direction and Entropy

The thermal dial t parameterises motion off the equator via the ratio $kT/(hH)$. This ratio connects mass scales to entropy, turning the (s, t)-plane into a bridge between dimensional analysis and thermodynamics.

Off the equator: thermal mixing

The general mass factorises as

$$m(s,t) = m(s,0)\left(\frac{kT}{hH}\right)^t. \tag{17}$$

The dimensionless ratio $kT/(hH) \sim 10^{29}$ compares thermal and cosmological–quantum energy scales at the present CMB temperature $T \approx 2.7\,\mathrm{K}$. Each unit of t shifts the mass by roughly 29 orders of magnitude.

Three energy scales

The ratio $kT/(hH)$ compares three forms of energy:

$$E = mc^2, \qquad E = hf, \qquad E \sim kT.$$

Inertial mass-energy mc^2 and quantum energy hf are exact identities. Thermal energy kT represents the characteristic energy per degree of freedom (up to factors of order unity). The Hubble parameter supplies a cosmological frequency $f \sim H$, so that hH represents the quantum of **action** accumulated over a Hubble time. The ratio $kT/(hH)$ therefore measures thermal energy against the cosmological quantum scale.

Action and a dimensionless entropy

Action has dimensions $[\mathcal{A}] = [E]\,[t] = \mathrm{J\,s}$. Since the Hubble time H^{-1} is the natural cosmic clock, the quantity

$$\mathcal{A}_U \equiv \frac{Mc^2}{H}$$

has dimensions of action: it is the inertial energy accumulated over one Hubble time. Dividing by Planck's constant produces a pure, dimensionless number:

$$\Sigma_{MI} \equiv \frac{\mathcal{A}_U}{h} = \frac{Mc^2}{hH}, \tag{18}$$

which may be interpreted as an entropy count analogous to S/k_B. The corresponding physical entropy is $S_{MI} = k_B\,\Sigma_{MI}$.

Numerically, $\Sigma_{MI} \sim 10^{120}$: the observable universe contains of order 10^{120} quanta of action when measured in units of h.

Since h has dimensions of angular momentum, $[h] = ML^2T^{-1}$, the same quantity admits an angular interpretation. For $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ with $\mathbf{p} = m\mathbf{v}$, we have $|L| \sim mrv$ and $[L] = ML^2T^{-1}$, identical to $[h]$. The dimensionless entropy count may thus be written

$$\Sigma_{MI} = \frac{\hat{L}_U}{h}, \qquad \hat{L}_U \equiv \frac{Mc^2}{H}.$$

Here $\hat{L}_U$ is an effective action scale (the inertial energy accumulated over one Hubble time) rather than evidence of literal global rotation.[1] For M fixed, $\Sigma_{MI} \propto H^{-1}$.

Length scales from $\{c, G, H, h\}$

A *horizon* is not a dimensional construct; it is a dynamical one. It arises whenever signals propagate at finite speed c while geometry evolves in time. In such a setting there is a natural causal boundary: beyond some distance scale, events cannot influence a given observer. Before assigning entropy to horizons, we ask the dimensional question: *what length scales can be built from the constants $\{c, G, H, h\}$?*

Seek $L \sim G^a\, h^b\, c^d\, H^e$. Using

$$[G] = M^{-1}L^3T^{-2}, \quad [h] = ML^2T^{-1}, \quad [c] = LT^{-1}, \quad [H] = T^{-1},$$

and requiring $[L] = L^1$, matching exponents gives

$$\begin{aligned} M&: \quad -a + b = 0, \\ T&: \quad -2a - b - d - e = 0, \\ L&: \quad 3a + 2b + d = 1. \end{aligned}$$

From M: $b = a$. Substituting: $5a + d = 1$ gives $d = 1 - 5a$, and $-3a - d - e = 0$ gives $e = 2a - 1$. The solution space has one free parameter a (nullity 1):

$$L(a) \sim G^a\, h^a\, c^{\,1-5a}\, H^{\,2a-1}. \tag{19}$$

Planck length. Setting H aside ($e = 0$) gives $2a - 1 = 0$, so $a = \frac{1}{2}$, yielding a purely quantum–gravitational length scale independent of cosmology:

$$\ell_P \sim \sqrt{\frac{Gh}{c^3}}.$$

Hubble length. Setting $a = 0$ gives a purely cosmological horizon scale:

$$R_H \sim \frac{c}{H}.$$

[1]For speculative extensions of this angular-momentum viewpoint, see [5].

Schwarzschild and Compton lengths. Introducing a free mass parameter M changes the dimensional problem. The Schwarzschild radius $r_s \sim GM/c^2$ is the scale at which gravitational binding energy becomes comparable to rest-mass energy Mc^2. The Compton wavelength $\lambda_C \sim h/(Mc)$ is the scale at which quantum energy hf balances rest-mass energy. Eliminating the mass parameter:

$$r_s\,\lambda_C \sim \frac{Gh}{c^3} \sim \ell_P^2.$$

The Planck length is the geometric mean of the Schwarzschild and Compton scales.

Area and horizon entropy

The product $r_s\,\lambda_C \sim Gh/c^3$ identifies the Planck area as the quantum–gravitational area scale. Working in $\hbar = h/(2\pi)$:

$$\ell_P^2 = \frac{\hbar G}{c^3}. \tag{20}$$

Checkpoint

Up to this point the argument has been purely dimensional: we have identified the admissible length and area scales. The further statement that entropy is *proportional to area* is not dimensional; it is a physical input from gravitational thermodynamics. The temperature normalisation that follows is also dynamical.

Accepting the Bekenstein–Hawking area law [6, 7], the entropy of a horizon with area A is

$$S = \frac{k_B\,A}{4\,\ell_P^2}.$$

The de Sitter horizon has proper area $A = 4\pi R_H^2$, with $R_H = c/H$. Hence

$$\frac{S_{dS}}{k_B} = \frac{4\pi\,(c/H)^2}{4\,\hbar G/c^3} = \frac{\pi c^5}{\hbar G H^2}. \tag{21}$$

Every factor is accounted for: 4π from the sphere, $\frac{1}{4}$ from Bekenstein–Hawking, together giving the prefactor π. Dimensional analysis forces H^{-2}; the normalisation is dynamical.

Numerically, $S_{dS}/k_B \sim 10^{122}$. Compare with $\Sigma_{MI} = Mc^2/(hH) \sim 10^{120}$: the two entropy counts differ by roughly two orders of magnitude, reflecting the distinction between action-based and area-based constructions.

The Bekenstein–Hawking formula treats the horizon as a thermodynamic object. If it carries entropy and satisfies a first-law relation $T dS = dU + p\,dV$, it must also carry a temperature. This temperature is what connects the horizon back to the (s,t)-plane: it fixes the value of $kT/(hH)$ at thermal equilibrium. For de Sitter space, the surface gravity is $\kappa = cH$, and horizon thermodynamics gives the Gibbons–Hawking temperature [8]

$$T_{dS} = \frac{\hbar H}{2\pi k_B},$$

arising from the Euclidean periodicity of de Sitter space in imaginary time.

At the present epoch, $T_{\mathrm{CMB}} \approx 2.7\,\mathrm{K}$ while $T_{dS} \sim 10^{-30}\,\mathrm{K}$. The ratio $T_{\mathrm{CMB}}/T_{dS} \sim 10^{30}$, giving $kT/(hH) \sim 10^{29}$ once the geometric factor $4\pi^2$ is included. Only in the hypothetical limit $T = T_{dS}$ which corresponds to a universe in thermal equilibrium with its cosmological horizon does

$$\left.\frac{kT}{hH}\right|_{T=T_{dS}} = \frac{1}{4\pi^2},$$

so that the mass family becomes

$$m(s,t)\big|_{T=T_{dS}} = c^{-2-5s} h^{1+s} H^{1+2s} G^{s} \left(\frac{1}{4\pi^2}\right)^{t}.$$

At $t = 0$ we sit on the thermal equator. Thermal equilibrium does not alter the s-structure; it merely rescales the mass along the t-direction by the fixed geometric factor $(4\pi^2)^{-t}$. The (s,t)-plane thus separates structural scaling (s) from thermodynamic displacement (t). The present universe is far from this regime.

Key insight

The (s,t)-plane separates structural scaling (s) from thermodynamic displacement (t)

The dimensionless action

$$\Sigma(s,t) = \frac{m(s,t)c^2}{hH}$$

inherits the same linear exponent structure as the mass family itself. In particular, the geometric mean property persists on the thermal equator:

$$\Sigma\big(-\tfrac{1}{2},0\big) = \sqrt{\Sigma(-1,0)\,\Sigma(0,0)}.$$

Numerically, $\Sigma(-1,0) \sim 10^{120}, \Sigma(0,0) \sim 10^{-16}$, so that the geometric mean of the Hubble entropy and the graviton entropy counts is the Planck action count:

$$\Sigma\big(-\tfrac{1}{2},0\big) \sim 10^{52},$$

The midpoint structure is therefore not a property of mass alone; it is a property of the exponent space itself. Mass, entropy, and action count are distinct projections of the same linear structure in (s,t). The (s,t)-plane is therefore not a bookkeeping device but the fundamental geometric object from which mass scales and entropy measures alike descend.

Dimensional analysis determines how entropy must scale with H; dynamics determines how much entropy there is. Once H is allowed to vary in time, the exponent geometry itself becomes dynamical.

Key insight

Two entropy constructions emerge from the same constants:

Quantity	Geometric origin	Scaling
$\Sigma_{MI} = Mc^2/(hH)$	volume $\times$ time (action)	$\propto H^{-1}$
$S_{dS}/k_B = \pi c^5/(G\hbar H^2)$	horizon area	$\propto H^{-2}$

Dimensional analysis fixes powers of H; horizon thermodynamics fixes normalisation.

On the thermal equator $t = 0$ the dimensionless action satisfies

$$\Sigma(s, 0) \propto H^{2s},$$

so specifying $H(t)$ determines the full time evolution. During radiation domination $H(t) \propto t^{-1}$, giving

$$\Sigma_{MI}(t) \propto t, \qquad S_{dS}(t) \propto t^2.$$

The different time dependences are therefore not additional assumptions but consequences of whether entropy arises from volume–time (action) or from area–geometry.

Conclusion

We asked: *how many masses can you build?* We arrived at the answer in stages. The spring gave us a fully determined system: one π-term, and that single combination was the answer. The density of the universe gave us another. The full six-constant mass problem is different in kind, since the target sits on the right-hand side and the nullspace supplies genuine freedom: rank four, nullity two, a two-parameter family,

$$m(s, t) = c^{-2-5s}\, h^{1+s-t}\, H^{1+2s-t}\, G^s\, (kT)^t,$$

containing the Planck, Hubble, Weinberg, and graviton masses as named coordinates on a thermal equator. The kT lock emerged from a single row of the matrix. The geometric mean structure revealed $m_P = \sqrt{m_\gamma \cdot m_H}$ as a consequence of the linearity of the exponent space.

But the paper's second result is that this same linear structure leads to entropy once the thermal direction is followed. The ratio $kT/(hH)$ that parameterises the t-axis generates the dimensionless action measure $\Sigma_{MI} = Mc^2/(hH) \sim 10^{120}$: the number of quanta of action in the observable universe. A separate dimensional construction, using the area ratio R_H^2/ℓ_P^2, produces the horizon entropy $S_{dS}/k_B = \pi c^5/(G\hbar H^2) \sim 10^{122}$. These two entropies scale differently with the Hubble parameter:

$$\Sigma_{MI} \propto H^{-1} \quad \text{(action: volume} \times \text{time)}, \qquad S_{dS}/k_B \propto H^{-2} \quad \text{(area: horizon geometry)}.$$

Dimensional analysis determines both scalings. Dynamics, through Euclidean periodicity and horizon thermodynamics, supplies the geometric prefactors and distinguishes volume–time from area–geometry.

Where on this map does Nature sit? That is a question for physics. But the map itself, from a 4×6 augmented matrix through Gaussian elimination to a two-parameter plane connecting mass scales and entropy, is pure linear algebra.

What dimensional analysis does

1. **Reveals structure.** A 4×6 linear system with rank 4 and nullity 2. Pivots, free variables, and row operations all carry physical meaning.

2. **Enforces the kT lock.** Temperature enters only through the energy kT, derived from a single row.

3. **Organises known scales.** Planck, Hubble, Weinberg, and graviton masses are coordinates in a continuous family, not isolated constructions.

4. **Exposes geometric mean relationships.** The linear exponent structure means $m_P = \sqrt{m_\gamma \cdot m_H}$, the Planck mass as midpoint.

5. **Connects mass to entropy.** The thermal direction generates $\Sigma_{MI} = Mc^2/(hH)$ and distinguishes action-based (H^{-1}) from area-based (H^{-2}) scalings.

What dimensional analysis does not do

1. **It does not predict.** Dimensional analysis maps the space of possibilities. Physics must select the point(s) that matter.

2. **It does not explain large numbers.** The 121-order span from m_γ to m_H is faithfully recorded as a distance in s-space; no mechanism is offered.

3. **It does not validate numerical coincidences.** That m_{234} is close to m_e or m_{24} to m_ν may be structural or accidental. Dimensional analysis cannot distinguish.

4. **It does not fix normalisations.** The factors of 2π, $3/(8\pi)$, and π that appear in oscillator periods, Friedmann density, and horizon entropy require dynamical input.

The broader lesson: Gaussian elimination is not just a computational technique. It is a tool for exposing the *geometry* of physical constraints.

An interactive companion exploring the entropy constructions and dimensional scalings discussed here is available at: `https://leelemacjames.github.io/dimensionEntropy/`

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Appendix: Some Geometric Means

The four fundamental masses on the equator (using $\{c, h, G, H\}$ without k or T) span **121 orders of magnitude**:

$$m_1 = \frac{c^3}{GH} \sim 1.8 \times 10^{53}\,\text{kg} \qquad (s = -1) \tag{22}$$

$$m_2 = \frac{hH}{c^2} \sim 1.6 \times 10^{-68}\,\text{kg} \qquad (s = 0) \tag{23}$$

$$m_3 = \left(\frac{h^3 H}{G^2}\right)^{1/5} \sim 4.3 \times 10^{-20}\,\text{kg} \qquad (s = -\tfrac{2}{5}) \tag{24}$$

$$m_4 = \sqrt{\frac{hc}{G}} \sim 5.5 \times 10^{-8}\,\text{kg} \qquad (s = -\tfrac{1}{2}) \tag{25}$$

Checkpoint

In the four-constant problem (c, h, G, H; no k or T), we have three base dimensions (M, L, T) and four constants, giving nullity $4 - 3 = 1$: a one-parameter family. This is the $t = 0$ equator of our full two-parameter family. There are $\binom{4}{3} = 4$ ways to choose three of the four constants and obtain a determined system. Each choice yields one of the four masses above. Allowing all four to participate gives the one-parameter family that *contains all four as special values of* s.

Scale	(s, t)	**Expression**	**Order (kg)**
Graviton	(0, 0)	hH/c^2	$\sim 10^{-68}$
Weinberg	$(-\frac{1}{3}$, 0)	$(h^2cH/G)^{1/3}$	$\sim 10^{-66}$
Planck	$(-\frac{1}{2}$, 0)	$\sqrt{hc/G}$	$\sim 10^{-8}$
Hubble	$(-1$, 0)	$c^3/(GH)$	$\sim 10^{52}$
Thermal rest	(0, 1)	kT/c^2	$\sim 10^{-40}$
Large-number	$(-1$, 1)	$c^3kT/(GH^2h)$	$\sim 10^{81}$

The geometric mean principle extends to higher-order means:

$$m_{234} = (m_2 \cdot m_3 \cdot m_4)^{1/3} = \left(\frac{h^{21}H^{12}}{c^{15}G^9}\right)^{1/30} \sim 3.4 \times 10^{-32}\,\text{kg},$$

within an order of magnitude of the electron mass $m_e = 9.1 \times 10^{-31}$ kg. Similarly,

$$m_{24} = \sqrt{m_2 \cdot m_4} = \left(\frac{H^2h^3}{Gc^3}\right)^{1/4} \sim 3 \times 10^{-38}\,\text{kg},$$

which is of the order of neutrino mass scales. These numerical proximities are noted without physical attribution; they arise from the arithmetic of the exponent space, not from any dynamical mechanism.

Appendix: Two Kinds of Large Number

Why this matters

The body of the paper declines to validate numerical coincidences, noting that the proximity of m_{234} to the electron mass may be structural or accidental. The invariant picture of Section 7 allows a sharper statement: it settles *in advance* which coincidences could possibly be structural, and it disposes of the most famous of them.

The classical coincidences are not on the map

The large numbers of Eddington [12] and Dirac [10, 11] are assembled from the very constants that span the (s, t)-plane, so one might expect them to be monomials $\Pi_0^a \Theta_T^b \alpha'^c$ in the invariants of Table 2. They are not. Fitting the standard coincidences to that form, with exponents ranging over half-integers, leaves residuals of six to ten orders of magnitude, which at these scales is no fit at all.

The obstruction is the one identified at the end of Section 7. Every classical coincidence contains a particle mass, and no particle mass lies on the map. Adjoining m_p supplies the fourth invariant $\alpha_G = Gm_p^2/(\hbar c)$ of Eq. (16), and in the enlarged space the coincidences stop being approximate and become exact identities. Three suffice.

The ratio of gravitational to electrostatic attraction between an electron and a proton:

$$\frac{F_{\text{grav}}}{F_{\text{EM}}} = \frac{\alpha_G}{\alpha} \frac{m_e}{m_p} \approx 4.41 \times 10^{-40}. \tag{26}$$

The age of the universe measured in proton Compton times:

$$\frac{m_p c^2}{\hbar H} = \sqrt{\frac{\alpha_G}{\Pi_0}} \approx 6.53 \times 10^{41}. \tag{27}$$

The Eddington number, the baryon count of a critical Hubble sphere [12]:

$$N_b = \frac{c^3}{2GHm_p} = \frac{1}{2} \frac{1}{\sqrt{\Pi_0\, \alpha_G}} \approx 5.52 \times 10^{79}. \tag{28}$$

Each has been verified to unit ratio against direct numerical evaluation. (Eqs. (27) and (28) use the $\hbar$-form $\Pi_0 = \hbar G H^2/c^5 \approx 1.39 \times 10^{-122}$; the h-form of Eq. (12) differs by 2π.)

The Dirac hypothesis, tested

Written in the invariant basis, Dirac's large-number hypothesis [10, 11] becomes a checkable relation rather than a comparison of numbers whose provenance is unclear. His $N_1 \sim N_2$, the claim that the age of the universe in atomic units matches the reciprocal of the gravitational-to-electromagnetic force ratio, is by Eqs. (26) and (27) the assertion

$$\sqrt{\frac{\alpha_G}{\Pi_0}} \sim \frac{\alpha}{\alpha_G} \frac{m_p}{m_e}, \qquad \text{that is,} \qquad \alpha_G^2 \sim \Pi_0,$$

up to factors of α and the mass ratio, of order 10^{-2} to 10^3 and negligible at this scale. The test is immediate:

$$\alpha_G^2 \approx 3.49 \times 10^{-77}, \qquad \Pi_0 \approx 1.39 \times 10^{-122},$$

a discrepancy of 45 orders of magnitude.

Checkpoint

The celebrated agreement holds between N_1 and N_2 as Dirac defined them, particular monomials in the invariants, and dissolves under change of basis. The coincidence is a statement about the exponents he chose rather than a relation between α_G and Π_0, and the time-variation of G inferred from it [11] does not follow. Modern bounds on $\dot{G}/G$ are consistent with zero [17].

First kind and second kind

Key insight

Invariants of the **first kind** are built from fundamental constants alone. There are three: Π_0, Θ_T and α', one per four-constant sub-basis, and they are the axes of the map.
Invariants of the **second kind** require a particle mass, which is a parameter of the matter content rather than a constant of nature. α_G is the first of these; every classical large number is built from it.

The distinction settles which numerical proximities could be structural. The 121-order equator span of Eq. (14) relates two first-kind quantities and is structural by construction. The proximity of m_{234} to m_e, or of m_{24} to the neutrino scale, compares a first-kind construction with a second-kind parameter, so no amount of dimensional analysis can make it structural: the exponent arithmetic generates the number, and the match to a measured mass carries no implication. This is the precise form of the caveat entered in the body of the paper. It applies equally to the horizon entropy of Section 8, which is first-kind and therefore derivable, and to the anthropic coincidences of Carr and Rees [15] and Barrow [16], which are second-kind and therefore not.

Quantity	In invariants	Value	Kind
Equator span	Π_0^{-1}	10^{121}	first
Horizon entropy	$\pi\Pi_0^{-1}$	10^{122}	first
Thermal displacement	Θ_T	10^{28}	first
Fine structure	α	10^{-2}	first
Force ratio	$(\alpha_G/\alpha)(m_e/m_p)$	10^{-40}	second
Age in Compton times	$\sqrt{\alpha_G/\Pi_0}$	10^{41}	second
Eddington number	$\frac{1}{2}(\Pi_0\alpha_G)^{-1/2}$	10^{79}	second

Table 3: Large numbers sorted by kind. First-kind quantities are derivable from constants; second-kind quantities require a measured particle mass and are therefore outside what dimensional analysis can predict.